

Proposition 2 - Envelope Theorem Derivation

1. Main objects

$$M(e; \theta, \eta, s) = p(se; \theta) \eta \Delta - c(e, \theta)$$

- $e \geq 0$: implementation effort supplied by the worker
- s : implementation skill
- θ : cognitive-tool quality
- $p(se; \theta)$: probability of successful implementation
- η : payoff judgment
- $\Delta > 0$: value of a successfully realized improvement
- $c(e; \theta)$: cost of implementation effort
- $M(e; \theta, \eta, s)$: expected net benefit from implementing an opportunity, conditional on having identified one

The worker chooses effort:

$$e^*(\theta, \eta, s) = \arg \max_{e \geq 0} M(e; \theta, \eta, s)$$

2. First-order condition

For an interior solution:

$$M_e = p_x(se; \theta) s \eta \Delta - c_e(e; \theta) = 0$$

So

$$p_x(se; \theta) s \eta \Delta = c_e(e; \theta)$$

↓
Marginal expected
benefit of effort

↓
Marginal cost
of effort

At the optimum:

$$M_e = 0$$

3. Optimized value

Define:

$$M^*(\theta, \eta, s) = M(e^*(\theta, \eta, s); \theta, \eta, s)$$

Where M^* is the maximum expected net benefit conditional on an opportunity after the worker has optimally chosen effort

Question:

How does M^* change when payoff judgement η increases?

By the chain rule:

$$\frac{dM^*}{d\eta} = M_\eta + M_e \frac{\partial e^*}{\partial \eta}$$

$\downarrow$ $\downarrow$
direct effect indirect effect
 through effort

Now:

$$M_\eta = p(se^*; \theta) \Delta$$

and from the FOC:

$$M_e = 0$$

Therefore

$$M_e \frac{\partial e^*}{\partial \eta} = 0$$

Hence:

$$\frac{\partial M}{\partial \eta} = M_\eta = p(se^*; \theta) \Delta$$

At the optimum, the first-order indirect effect through the change in optimal effort disappears.

Effect of Payoff Judgement on Tool Adoption

4. Total value of the iterative improvement process

Let

$$V_0(\theta) = \Gamma M^*(\theta)$$

where:

$$\Gamma = \sum_{t=0}^{\infty} \left(\prod_{i=0}^t p(i) \right) \delta^t$$

- $p(t)$: opportunity judgement in iteration t , represented by the probability of recognizing another improvement opportunity
- δ : discount factor
- Γ : opportunity multiplier; captures how often future opportunities are discovered and how much are valued
- V_0 : expected discounted value of the entire iterative improvement process

5. Gain from adopting the cognitive tool

Define

$$D(\eta) = V_0(1) - V_0(0)$$

↓
Gain from
tool adoption

Since:

$$V_0(\theta) = \Gamma \pi^*(\theta)$$

then

$$D(\eta) = \Gamma [\pi^*(1, \eta) - \pi^*(0, \eta)]$$

Differentiate with respect the payoff judgement:

$$\frac{dD(\eta)}{d\eta} = \Gamma \left[\frac{\partial \pi^*(1)}{\partial \eta} - \frac{\partial \pi^*(0)}{\partial \eta} \right]$$

Using the envelope theorem:

$$\frac{\partial \pi^*(1)}{\partial \eta} = p(se^*(1); 1) \Delta$$

$$\frac{\partial \pi^*(0)}{\partial \eta} = p(se^*(0); 0) \Delta$$

Therefore:

$$\frac{dD}{d\eta} = \Gamma \Delta [p(se^*(1); 1) - p(se^*(0); 0)]$$

6. Economic intuition

There are two opposing effects of the cognitive tool

i) Direct technological effect:

$$\theta \uparrow \rightarrow p(se; \theta) \uparrow$$

The tool makes implementation easier or more successful

ii) Behavioral effort effect:

$$\theta \uparrow \rightarrow e^* \downarrow \rightarrow p(se^*; \theta) \downarrow$$

The worker optimally reduces implementation effort

Therefore:

Payoff judgement is only conditionally complementary to the tool

More precisely:

$$\frac{dD}{dn} > 0$$

if and only if:

$$p(cse^*(c); 1) > p(cse^*(c); 0)$$

Final intuition:

Payoff judgement makes tool adoption more valuable only when equilibrium implementation success is higher with the tool than without it.